

Staff Cloud Engineer (AWS) — Take-Home Exercise

Skylo Regional Hub: AWS Architecture Design

Time budget: 90 minutes. Hard stop. We mean it — we would rather see sharp reasoning with acknowledged gaps than a polished document that took you a weekend. If you run out of time, note what you'd do next and submit.

1. Context

Skylo runs a global Non-Terrestrial Network: smartphones and IoT devices connect directly over satellites, served by our cloud-native 3GPP vRAN and core. We're expanding to a new continent and need a **Regional Hub in us-west-2** with two functions:

1. **Data ingress** — high-volume, real-time connectivity data from multiple satellite ground stations, arriving over AWS Direct Connect.
2. **Core network processing** — containerized 3GPP core components managing device sessions, processing data, routing traffic.

Constraints that matter: resilient to single-AZ failure; compute scales with traffic; least-privilege and zero-trust; SOC 2 in scope; everything provisioned as IaC; must connect to Direct Connect and to other accounts in the Skylo AWS Organization.

Assume anything not specified. State your assumptions.

2. Deliverables

Two files. Submit as a zip or a private Git link.

Deliverable A — **DESIGN.md** (~60 min, 2–3 pages)

Prose, bullets, and one diagram. ASCII, a photo of a whiteboard, or a linked image — all fine.

Do not spend time on diagram tooling.

A1. Network (~20 min)

- Diagram the VPC: subnet tiers, route tables, NAT, and the Transit Gateway attachment carrying Direct Connect and east-west Org traffic.
- Your CIDR strategy, and why. Include how it avoids collisions as Skylo adds more regional hubs.

A2. Compute (~10 min)

- **EKS or ECS?** Pick one. Justify it against scalability, operational overhead, and the specifics of running 3GPP core workloads.
- Name the one thing that would make you reverse this decision.

A3. Storage (~10 min)

Pick a service and give a one-line justification for each:

- Short-term session state — high throughput, low latency.
- Long-term archival of connection logs.

A4. HA and DR (~10 min)

- How the design survives an AZ loss.
- Your strategy if **all of us-west-2** is gone. Name your target RTO/RPO and what that costs.

A5. Security and observability (~10 min)

Keep this tight — bullets are fine:

- The IAM role model for the workloads (e.g. IRSA), and the permissions you'd deliberately **withhold**.
- The **first 3 AWS security services** you'd turn on, and why those three first.
- The **top 3 metrics** you'd alert on for the containerized workloads.
- Your logging/monitoring/alerting stack in one or two sentences (CloudWatch, Datadog, Prometheus, ELK — your call).

Deliverable B — **vpc.tf** (~30 min, one file)

A single annotated Terraform file for the **foundational network only**: VPC, subnets, route tables, IGW, NAT, and the Transit Gateway attachment (the TGW itself can be a **data** source or a variable — assume it exists).

- It does **not** need to **terraform plan** cleanly. It does **not** need to be complete.

- **Do not build a module tree.** Instead, add a short comment block at the top sketching how this would be decomposed into modules in the real repo.
 - Comments explaining *why* count more than working HCL.
 - Skip EKS/ECS resources entirely — just leave a commented stub showing where the cluster module would be called.
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3. How we evaluate

Signal	What we're reading for
Staff-level judgment	Explicit trade-offs on cost, complexity, blast radius. Decisions, not surveys of options.
AWS depth	Networking, compute, and security choices that reflect real operating experience.
IaC instinct	Structure, naming, and boundaries — visible even in one file.
Security by default	Built into A1–A4, not confined to A5.
Communication	Could a peer implement from this document? Could a VP follow the summary?

What we are not scoring: diagram polish, `terraform validate` passing, exhaustive service coverage, prose length.

4. Follow-up

We'll spend the first 30 minutes of the technical interview on this design — pushing on your trade-offs and changing a requirement to see how the architecture bends. Come ready to defend it and to revise it.

Questions before you start? Reply to the recruiter thread.